Partial Pooling Across Hosts: A Multilevel Model for Airbnb Listing Prices

Statistical Rethinking, Chapter 13 — Worked Exercise

TalRamot46*

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Abstract

This essay applies the varying-intercepts multilevel model of Chapter 13 of *Statistical Rethinking* to [data set]. The outcome of interest is [outcome], grouped by [cluster variable]. We fit [model] using Hamiltonian Monte Carlo and compare the resulting cluster-level estimates to their no-pooling and complete-pooling counterparts. We find [headline result], which illustrates [the shrinkage / regularisation lesson].

Contents

*Replace with your full name, ID, and course details.

1 Introduction and Research Question

Chapter 13 of *Statistical Rethinking* introduces multilevel (hierarchical) models as a principled compromise between two extremes: *complete pooling*, which ignores cluster identity entirely, and *no pooling*, which estimates each cluster in isolation. [Extend this motivation to your own setting.]

The data used here are [describe source]. Each row is a single listing, but listings are nested within hosts: a single host may operate many listings, and those listings plausibly share a common pricing strategy. Treating the rows as independent would therefore overstate the effective sample size and understate uncertainty.

Our research question is:

> [State it as a question about $\bar{\alpha}$, σ_{host}, or the ratio between them.]

The remainder of the essay proceeds as follows. Section 2 describes the data and preprocessing. Section 3 states the statistical model and justifies each prior. Section 4 reports the fitting procedure and diagnostics. Section 5 presents the posterior. Section 6 examines shrinkage directly, and Section 8 concludes.

2 Data

2.1 Source and Variables

The raw file `listings.csv` contains [N_{raw}] listings and 91 columns. The variables used in this analysis are summarised in Table 1.

Table 1: Variables used in the analysis.

Variable	Type	Description
`host_id`	integer	Identifier of the host; defines the cluster.
`price`	string	Nightly price as scraped, e.g. `$120.00`.
`review_scores_rating`	real	Mean guest rating on a 0–5 scale.
`number_of_reviews`	integer	Count of reviews; used as a filter.
`price_clean`	real	*Derived.* `price` parsed to a number.
`log_price`	real	*Derived.* log(`price_clean`); the outcome.
`host_idx`	integer	*Derived.* Contiguous index $1, \ldots, J$ over hosts.

2.2 Preprocessing

Three preprocessing choices deserve comment. First, listings with no reviews are dropped, since [reason]. Second, price is modelled on the log scale because [reason]; this also guarantees that the Gaussian likelihood places no mass on negative prices once exponentiated. Third, the data are subsampled to $n = 2000$ rows to keep sampling times manageable, at the cost of [reason].

Cluster structure. After filtering, there are $J = [J]$ distinct hosts across $n = 2000$ listings, a mean of $[n/J]$ listings per host. [Report the distribution — how many hosts contribute exactly one listing? Those are the clusters that shrink the most.]

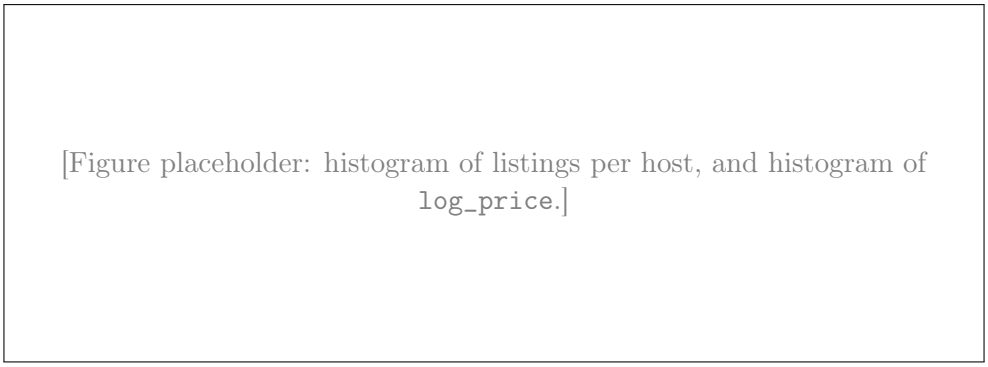

Figure 1: Distribution of listings per host (left) and of `log_price` (right).

3 The Model

3.1 Statistical Specification

Let y_i denote `log_price` for listing $i = 1, \ldots, n$, and let $j[i] \in \{1, \ldots, J\}$ denote the host of listing i. The varying-intercepts model is

$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i,\ \sigma) && && (1)\\
\mu_i &= \alpha_{j[i]} && && (2)\\
\alpha_j &\sim \text{Normal}(\bar{\alpha},\ \sigma_{\text{host}}) && \text{[adaptive prior]} && (3)\\
\bar{\alpha} &\sim \text{Normal}(4.5,\ 1) && \text{[hyper-prior]} && (4)\\
\sigma_{\text{host}} &\sim \text{Exponential}(1) && \text{[hyper-prior]} && (5)\\
\sigma &\sim \text{Exponential}(1). && && (6)
\end{aligned}
$$

The third line is what makes this a multilevel model. The prior for each host intercept α_j is not fixed in advance but is itself estimated from the data through $\bar{\alpha}$ and σ_{host}. Hosts therefore *learn from one another*: a host with a single listing is pulled towards the population mean $\bar{\alpha}$, whereas a host with many listings largely retains its own estimate.

3.2 Prior Justification

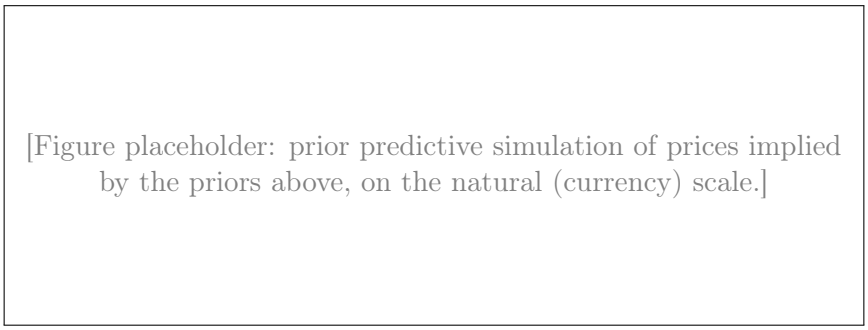

Figure 2: Prior predictive check. Priors should permit plausible prices without concentrating mass on absurd ones.

3.3 Comparison Models

- **Complete pooling:** $\mu_i = \alpha$, one intercept for all listings.

- **No pooling:** $\mu_i = \alpha_{j[i]}$ with $\alpha_j \sim \text{Normal}(4.5, 1)$ fixed — each host estimated independently.
- **Partial pooling:** the model above.

Compare them with `compare(m1, m2, m3, func=WAIC)` and discuss the effective number of parameters, which is the clearest quantitative signature of regularisation.

4 Fitting and Diagnostics

The model was fit with `ulam()`, which compiles the formula to Stan and samples via Hamiltonian Monte Carlo: 4 chains, 2000 iterations each (half warm-up). A `brms` implementation of the same structure is included for comparison.

4.1 Diagnostics

Table 2: Convergence diagnostics for the population-level parameters.

Parameter	Mean	SD	ESS	$\hat{R}$
$\bar{\alpha}$	–	–	–	–
σ_{host}	–	–	–	–
σ	–	–	–	–

5 Results

5.1 Population-Level Parameters

5.2 Variance Decomposition

$$\text{ICC} = \frac{\sigma^2_{\text{host}}}{\sigma^2_{\text{host}} + \sigma^2} = [\cdot] \quad (89\%\ \text{CI} : [\cdot, \cdot]). \tag{7}$$

5.3 Host-Level Intercepts

[Figure placeholder: caterpillar plot of the α_j posteriors, ordered by mean, with $\bar{\alpha}$ as a horizontal reference line.]

Figure 3: Posterior means and 89% intervals for host intercepts α_j.

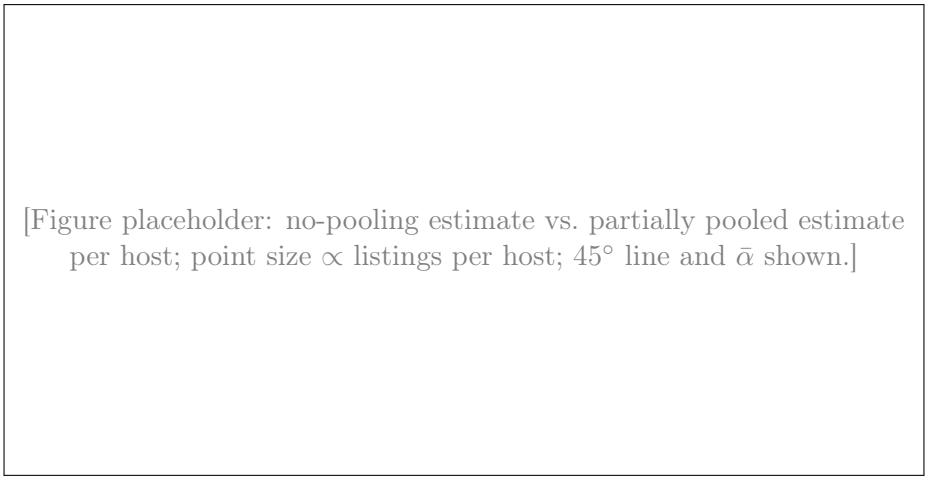

Figure 4: Shrinkage of host-level estimates towards the population mean.

6 Shrinkage and Partial Pooling

7 Limitations

8 Conclusion

A R Code

A.1 Multilevel Model via `rethinking::ulam`

```r
library(tidyverse)
library(rethinking)

# 1. Load and clean data
df <- read_csv("listings.csv") %>%
  # Select required columns (review_scores_rating exists in this detailed
      dataset)
  select(host_id, price, review_scores_rating, number_of_reviews) %>%
  filter(!is.na(price), !is.na(review_scores_rating), number_of_reviews > 0)
      %>%
  mutate(
    # Clean price: remove '$' and ',' then parse to numeric
    price_clean = parse_number(as.character(price)),
    log_price   = log(price_clean)
  ) %>%
  filter(price_clean > 0) %>%
  slice_sample(n = 2000)

# 2. Rethinking requires CLEAN integer index numbers (1, 2, 3, ... N)
df$host_idx <- as.integer(as.factor(df$host_id))

# 3. Create list for ulam
dat <- list(
  log_price = df$log_price,
  host_idx  = df$host_idx
)

# 4. Chapter 13 Multilevel Model Formula
f_multilevel <- alist(
  log_price ~ dnorm(mu, sigma),

```

```r
  # Linear model: unique intercept 'a' per host
  mu <- a[host_idx],

  # Adaptive Prior (Varying Intercepts / Partial Pooling)
  a[host_idx] ~ dnorm(a_bar, sigma_host),

  # Hyper-priors
  a_bar ~ dnorm(4.5, 1),
  sigma_host ~ dexp(1),

  # Residual variation
  sigma ~ dexp(1)
)

# 5. Fit model via ulam (Stan HMC engine)
m_rethinking <- ulam(
  f_multilevel,
  data = dat,
  chains = 4,
  cores = 4,
  iter = 2000
)

# 6. View estimates (depth = 2 prints varying host intercepts alongside a_bar and sigma)
sink("precis_output.txt")
precis(m_rethinking, depth = 2) # or depth = 1
sink()
```

Listing 1: `rethinking_code.R` — varying intercepts on `log_price` by host.

A.2 Equivalent Model via `brms`

```r
library(tidyverse)
library(brms)

# 1. Load and clean data
# Make sure "listings.csv" is in your working directory, or specify the full path
df <- read_csv("listings.csv") %>%
  select(host_profile_id, review_scores_rating, number_of_reviews) %>%
  filter(!is.na(review_scores_rating), number_of_reviews > 0) %>%
  # Ensure host_profile_id is treated as a categorical factor for varying intercepts
  mutate(host_profile_id = as.factor(host_profile_id)) %>%
  # Filter down to top N rows to keep sampling quick
  slice_sample(n = 2000)

# 2. Set Bayesian Priors
# Chapter 13 focuses on partial pooling via varying intercepts (1 | host_id)
priors <- c(
  prior(normal(4.5, 0.5), class = Intercept), # Prior on average rating
  prior(exponential(1), class = sd),            # Prior on standard deviation across hosts
  prior(exponential(1), class = sigma)         # Prior on residual variation within hosts
)

# 3. Fit Varying Intercept Model via brms (uses Stan backend under the hood)
m_partial_pooling <- brm(
  formula = review_scores_rating ~ 1 + (1 | host_profile_id),
  data = df,
```

```
  family = gaussian(),
  prior = priors,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000
)

# 4. View Model Summary
summary(m_partial_pooling)
```

Listing 2: `code.R` — the same varying-intercepts structure in `brms` syntax.

A.3 Reproducibility

References

[1] McElreath, R. (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*, 2nd ed. CRC Press. Chapter 13.

[2] Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, 80(1), 1–28.

[3] Stan Development Team (2024). *Stan Modeling Language Users Guide and Reference Manual*.

[4] Inside Airbnb. `http://insideairbnb.com/get-the-data/`. Accessed [date].